PROJECT 3

Deadline:	Submit by midnight 15th of October 2025
Evaluation:	20% of your final course grade.
Late Submission:	**No late submissions accepted since this is the of the semester.**
Work	This assignment should be done in **groups** made up of **three** to **four** students. One person is not a group so please don't ask if you can work alone.
Purpose:	Learning outcomes 1 - 5 from the course outline.

You are expected to come up with topics for your group at the earliest possible stage so that you can commence work on development. Preferably, discuss your chosen topic and what it is you plan to develop with the teaching staff before commencing work.

Groups must be formed by **September 12** and the class Google Sheet listing your members and proposed topic must be filled out. The Google Sheet is here: https://docs.google.com/spreadsheets/d/1sGA9W6XENIPgrN9kKbGVijj6hFjwmO7EjXyu12PJc8U/edit?usp=sharing

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*** Plagiarism***

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It is mandatory that any assessment items that you submit during your University study are your own work. Massey University takes a firm stance on academic misconduct, such as plagiarism and any form of cheating.

Plagiarism is the copying or paraphrasing of another person's work, whether published or unpublished, without clearly acknowledging it. It includes copying the work of other students and reusing work previously submitted by yourself for another course. **It also includes the copying of code from unacknowledged sources**.

Academic integrity breaches impact on students as it disadvantages honest students and undermines the credibility of your qualification. Plagiarism, and cheating in tests and exams will be penalised; it is likely to lead to loss of marks for that item of assessment and may lead to an automatic failing grade for the course and/or exclusion from reenrolment at the University.

Please see the Academic Integrity Guide for Students on the University website for more information. The Guide steps you through the University Academic Integrity Policy and Procedures. For example, you will find definitions of academic integrity misconduct, such as plagiarism; how misconduct is determined and managed; and where to find resources and assistance to help develop the skills of academic writing, exam preparation and time management. These skills will help you approach university study with academic integrity.

PROJECT OUTLINE:

This final project for 297.301 is an extension of the third project in 297.201.

You are again asked to create a data science-related software product based on a chosen real-world problem domain/dataset which showcases predictive analytics. You are strongly encouraged again to develop machine learning predictive or forecasting models which are deployed and demonstrated through some software artefact or online web app. The recommendation is that you use Streamlit as a front-end for your model outputs but you can of course use some other web app or technology that you are familiar with too.

You need to explore and come up with a domain/dataset for your project and analyse its ability to support the type of analysis and predictive analytics you wish to conduct. Check with the teaching team if your topic is suitable before committing to it.

PROJECT REQUIREMENTS

1. Presentation: each group will present their work and demo with a max time of 15 minutes
2. Originality: ensure your work is as unique as possible by combining multiple datasets in order to make a unique contribution
3. Deliverable: your submission should have two components:
    a. The code for the software artefact/web app
    b. Jupyter notebook presented in a professional formal like a report; highlighting all the experimentation, interpretation, hyperparameter tuning, use of XAI etc.

SUBMISSION REQUIREMENTS AND EVALUATION

Make sure you do these four things:

1. One submission per team
2. Submit a separate document (or include this at the top of a notebook) that details what each team member contributed to the assignment. Not all contributors will be awarded the same mark. Each team member must submit their own version of how each team member contributed.
3. Each member of the class will be marked individually – everyone must take part in the presentation. **Students enrolled to study on-campus in Auckland must deliver their presentations in-person. Distance students can of course either make a recording of their presentations, or present live online during our final week 12 session from 1pm-4pm on Thursday.**
4. Everyone must watch and mark others' presentations and submit the scores

MARKING CRITERIA:

Marks will be awarded for different components of the project using the following rubric:

Component	**Requirements**	**Marks**
Project presentation not exceeding 15 minutes	- Marking rubric for the presentations will be posted on Stream.	20%
Productionised deliverable:	- Submission of all your web app and other supporting code defining your software artefact - The manner in which you have showcased your predictive analytics work. - Use of AWS Services where possible especially in accessing deployed models through AWS APIs. Discuss how you have used AWS services where relevant.	25%
Jupyter Notebook Professional report	- A single professional and structured Jupyter notebook (pretend that you are submitting this to a potential client) - Include some EDA where needed, but focus primarily on robust experimentation with feature engineering, a variety of algorithms and their tuning, XAI, and providing rich analysis and interpretation at each step. - Make sure that you use a rich array of evaluation metrics and present all your modelling and accuracies from different approaches effectively.	35%
Originality, creativity and difficulty of the project	- Explain how and in which aspects is your work novel and differs from other works. What is its contribution and where did most of the difficulty lie?	15%
Submission of team member contribution document (every team member must submit their own version) **and** the submission of the class presentation scores.	- Submission of documents outlining team contributions - Submission of team presentation marks	5%

Hand-in: **Zip**-up all your **notebook (include an HTML version), python and other application source files** into a single file. Submit this file via **stream.**

If you have any questions or concerns about this assignment, please ask the lecturer sooner rather than closer to the submission deadline.

It goes without saying, that using GenAI to write assignment solutions for you is not acceptable and constitutes academic dishonesty and cheating. However, you are permitted to use GenAI for streamlit code generation since this has not been a taught component of the course.